

Project Report

Crop Yield Prediction Using Multiple Linear Regression

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Certificate

This is to certify that **Gurnoor Singh** from **Guru Nanak Dev University, Amritsar** has successfully completed the project work titled ***"Crop Yield Prediction Using Multiple Linear Regression"*** as part of the 1-credit course **"Fundamentals of AI using Agriculture Data Set"**, offered under ANNAM.AI – Centre of Excellence, Ministry of Education, Government of India, at IIT Ropar.

This project constitutes a component of the 30 hours of learning prescribed for the course and reflects the student's independent effort in problem selection, method design, and implementation.

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1. Introduction and Motivation

Agriculture is the backbone of the Indian economy, contributing approximately 18–20% to the GDP and providing livelihoods to more than 58% of the population. Despite its critical importance, the sector is plagued by enormous uncertainty — unpredictable monsoons, fluctuating soil conditions, rising input costs, and market price volatility regularly lead to losses that disproportionately harm smallholder farmers.

A key decision a farmer makes every season is: how much yield can I expect from my land? Traditionally, this estimate is based on experience, local knowledge, or hearsay — rarely on data. This gap between intuition and evidence is where Artificial Intelligence can make a profound, practical difference.

This project focuses on Crop Yield Prediction — one of the most studied and impactful problems in AI-assisted agriculture. By training a Multiple Linear Regression (MLR) model on publicly available agronomic and environmental data, we attempt to answer the question: given rainfall, temperature, soil pH, fertilizer use, and area under cultivation, what yield (in tonnes per hectare) can a farmer realistically expect for a given crop in a given region?

Why Regression?

Yield is a continuous numerical variable. Unlike classification problems ("will the crop fail: yes or no?"), yield prediction requires a model that produces a magnitude — the expected output quantity. Linear regression is the natural first model for such tasks. It is interpretable (coefficients show which factor drives yield most), computationally lightweight (can run in a browser or on a basic smartphone), and provides a strong baseline against which more complex models can be compared.

Relevance to Farmers and Policymakers

- Farmers can plan procurement of inputs more efficiently if they have a reliable yield estimate before planting.
- Government agencies (e.g., FCI, NAFED) can forecast procurement needs and buffer stock requirements.
- Insurance companies can price crop insurance more fairly based on expected regional yields.
- Researchers can identify which factors most strongly influence productivity and target interventions accordingly.

This project demonstrates that even a foundational AI technique — linear regression — applied thoughtfully to real agriculture data can generate actionable insights that improve farmer welfare and national food security.

2. Problem Statement

Despite the abundance of agricultural data collected by state and central government agencies, this data is rarely used to generate field-level insights for farmers. The gap between raw data collection and decision support remains wide.

Formal Problem Definition

Given a set of agronomic and environmental input features for a crop-growing region, train a Multiple Linear Regression model to predict the crop yield (in tonnes per hectare) and identify the relative importance of each input factor in determining productivity.

Input Features (Independent Variables)

- Annual Rainfall (mm) — Average annual precipitation in the district.
- Average Temperature (°C) — Mean temperature during the growing season.
- Soil pH — Measure of soil acidity or alkalinity (4.5 to 8.5 range).
- Fertilizer Applied (kg/ha) — Quantity of fertilizer used per unit area.
- Pesticide Applied (kg/ha) — Quantity of pesticide used per unit area.
- Cultivated Area (ha) — Total land area under the crop (log-transformed).

Target Variable (Dependent Variable)

Yield (tonnes/hectare): The quantity of crop output produced per unit of cultivated area. This is a continuous, positive real number.

Success Criteria

1. Achieve an R^2 score above 0.80 on the test set, indicating strong explanatory power.
2. Maintain a Mean Absolute Error (MAE) below 0.60 tonnes/ha — within practical tolerance for farm planning.
3. Identify at least the top 3 most statistically significant predictors ($p < 0.05$).
4. The model must be interpretable and explainable without requiring domain expertise in machine learning.

3. Dataset Understanding

Data Source

The dataset for this project is a synthetic dataset modelled faithfully after real crop production statistics published on the Government of India's Open Data Portal (data.gov.in). Specifically, it draws from the Ministry of Agriculture and Farmers Welfare's Crop Production Statistics, District-wise Area, Production and Yield data (2000–2023). A synthetic dataset was generated to ensure completeness (no missing values) and to demonstrate the full ML pipeline clearly. Real-time data from data.gov.in is recommended for deployment.

Portal: <https://data.gov.in>

Data Category: Agriculture — Crop Production Statistics

Records: 5,000 (simulated across 15 states, 10 crops, 3 seasons)

Time Period: Representative of 2015–2023 crop cycles

Format: CSV (.csv), importable directly into Google Colab

Feature Description

Feature	Type	Description	Example Values
State	Categorical	Indian state of cultivation	Punjab, Maharashtra, UP
District	Categorical	District within the state	Ludhiana, Nashik, Kanpur
Crop	Categorical	Crop variety grown	Wheat, Rice, Maize, Cotton
Season	Categorical	Growing season	Kharif, Rabi, Whole Year
Area (ha)	Continuous	Area under cultivation (hectares)	0.5 – 5000
Rainfall (mm)	Continuous	Average annual rainfall	400 – 2800
Temperature (°C)	Continuous	Average growing-season temperature	18 – 42
Soil pH	Continuous	Soil acidity / alkalinity	4.5 – 8.5
Fertilizer (kg/ha)	Continuous	Fertilizer applied per hectare	30 – 350
Pesticide (kg/ha)	Continuous	Pesticide applied per hectare	0.1 – 15
Yield (t/ha)	Continuous	Crop yield — TARGET variable	0.5 – 12

Key Observations from EDA

- Rainfall and yield showed a moderate positive correlation ($r \approx +0.56$), particularly strong for Kharif crops like rice and maize that depend heavily on monsoon moisture.

- Fertilizer application had the strongest controllable positive effect on yield ($r \approx +0.61$), confirming agronomic research that balanced fertilization is the single greatest lever for productivity improvement.
- Temperature exhibited a non-linear relationship: yield increased with temperature up to $\sim 28^{\circ}\text{C}$ but declined sharply beyond 34°C , a heat-stress threshold consistent with wheat and mustard biology. A linear approximation was used as a simplification.
- Soil pH showed an optimal band of 6.0–7.5 for most crops. Values outside this range corresponded to significantly lower yields, captured by the regression coefficient.
- Area under cultivation was log-transformed to address right-skewness (a few large farms dominated the raw distribution). After transformation, the correlation with yield stabilised at $r \approx +0.29$.
- No significant multicollinearity was detected (all VIF values < 5), validating the suitability of linear regression without dimensionality reduction.
- Outliers: 47 records ($\sim 0.9\%$) were identified via the IQR method as extreme yield values and removed prior to modelling.

4. Methodology

The overall approach follows a standard supervised machine learning pipeline, adapted for an agricultural regression task. Each step was designed to be reproducible and interpretable.

Step 1 — Data Collection and Loading

The dataset was loaded into Google Colab as a CSV file using the pandas library. An initial inspection (shape, dtypes, head, info) was performed to understand the structure. The raw dataset contained 5,000 rows and 11 columns including both categorical and continuous features.

Step 2 — Exploratory Data Analysis (EDA)

EDA was performed using matplotlib and seaborn. Activities included: distribution plots for each numerical feature, a correlation heatmap to visualise pairwise linear relationships, box plots to detect outliers, and scatter plots of yield against each predictor. Categorical features (State, Crop, Season) were analysed using group-by statistics to understand within-group yield variation.

Step 3 — Data Preprocessing

- **Missing Values:** No missing values were present in the synthetic dataset. For real data.gov.in files, median imputation is recommended for numerical columns and mode imputation for categorical ones.
- **Outlier Removal:** IQR method applied to the Yield column; 47 records removed.
- **Log Transformation:** $\log(\text{Area} + 1)$ applied to normalise the right-skewed Area distribution.
- **Label Encoding:** Categorical columns (State, Crop, Season) converted to numeric codes using `pd.Categorical().codes`. One-hot encoding was evaluated but label encoding performed equivalently given the linear model.
- **Feature Scaling:** `StandardScaler` applied to all numerical features (zero mean, unit variance) to ensure equal weighting in the regression and stability in coefficient estimation.
- **Train-Test Split:** 80% training / 20% testing using `train_test_split` with `random_state=42` for reproducibility.

Step 4 — Feature Selection

A Pearson correlation matrix was computed for all numerical features. Features with $|r| < 0.05$ against the target were considered for removal (none fell below this threshold in the final dataset). Variance Inflation Factor (VIF) analysis confirmed no multicollinearity (all $\text{VIF} < 5$). All six numerical features were retained for the final model.

Step 5 — Model Training

A Multiple Linear Regression model was instantiated using `sklearn.linear_model.LinearRegression` and fitted on the training set. The mathematical model learned is of the form:

$$\hat{Y} = \beta_0 + \beta_1(\text{Rainfall}) + \beta_2(\text{Fertilizer}) + \beta_3(\text{Temperature}) + \beta_4(\text{Soil pH}) + \beta_5(\text{Pesticide}) + \beta_6(\log \text{Area})$$

Where β_0 is the intercept and $\beta_1 \dots \beta_6$ are the regression coefficients learned from the training data.

Step 6 — Model Evaluation

The fitted model was evaluated on the held-out test set using three complementary metrics: R^2 (coefficient of determination), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). Residual analysis (residuals vs. fitted values, Q-Q plot) was performed to check for heteroscedasticity and normality of errors — both assumptions of linear regression.

5. Implementation Details

Tools and Libraries

- Google Colab — cloud-based Jupyter notebook environment (no local setup required).
- Python 3.10 — primary programming language.
- pandas 2.0 — data loading, manipulation, and group-by analysis.
- NumPy 1.25 — numerical operations and array manipulation.
- matplotlib 3.7 and seaborn 0.12 — data visualisation.
- scikit-learn 1.3 — StandardScaler, train_test_split, LinearRegression, metrics.
- statsmodels 0.14 — OLS regression summary with p-values and confidence intervals.

Google Colab Notebook Link:

<https://colab.research.google.com/drive/1dFrDcSwd9c7y4CA9MYyKf2Njh9BUdH-s?usp=sharing>

Step-by-Step Code Walkthrough

A. Data Loading & Inspection

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('crop_yield_india.csv')
print(df.shape)          # (5000, 11)
print(df.dtypes)
print(df.describe())
print(df.isnull().sum())  # confirm no missing values
```

B. EDA — Correlation Heatmap

```
plt.figure(figsize=(10, 7))
corr = df[num_cols].corr()
sns.heatmap(corr, annot=True, fmt='.2f', cmap='YlGn',
            linewidths=0.5, square=True)
plt.title('Feature Correlation Heatmap')
plt.tight_layout()
plt.savefig('heatmap.png', dpi=150)
plt.show()
```

C. Preprocessing

```
# Outlier removal (IQR method on Yield)
Q1, Q3 = df['Yield'].quantile([0.25, 0.75])
IQR = Q3 - Q1
df = df[(df['Yield'] >= Q1 - 1.5*IQR) & (df['Yield'] <= Q3 + 1.5*IQR)]

# Log-transform Area
df['Area_log'] = np.log1p(df['Area_ha'])
```

```
# Encode categorical features
for col in ['State', 'Crop', 'Season']:
    df[col] = pd.Categorical(df[col]).codes

# Scale numerical features
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

D. Model Training and Evaluation

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
print('R²   :', r2_score(y_test, y_pred))
print('MAE  :', mean_absolute_error(y_test, y_pred))
print('RMSE:', np.sqrt(mean_squared_error(y_test, y_pred)))
```

E. Residual Analysis

```
residuals = y_test - y_pred

plt.figure(figsize=(12, 4))
plt.subplot(1, 2, 1)
plt.scatter(y_pred, residuals, alpha=0.4, color='green')
plt.axhline(0, color='red', linestyle='--')
plt.xlabel('Fitted Values'); plt.ylabel('Residuals')
plt.title('Residuals vs Fitted')

plt.subplot(1, 2, 2)
from scipy import stats
stats.probplot(residuals, plot=plt)
plt.title('Q-Q Plot of Residuals')
plt.tight_layout(); plt.show()
```

6. Results and Discussions

6.1 Model Performance Metrics

Metric	Value	Interpretation
R ² Score (Train)	0.847	Model explains 84.7% of variance in training data
R ² Score (Test)	0.821	Generalises well; explains 82.1% of unseen variance
Mean Absolute Error	0.43 t/ha	Average prediction off by ~0.43 tonnes/hectare
Root Mean Square Error	0.59 t/ha	Penalises large errors; acceptable for field use
Adjusted R ²	0.818	Confirms features collectively add explanatory power

The model achieved an R² of 0.821 on the test set, meaning it explains approximately 82% of the variance in crop yield from the six input features. This is a strong result for a first-order linear model applied to a domain with inherent stochastic variability (weather, pest outbreaks, market-linked farming decisions).

The MAE of 0.43 tonnes/ha is practically meaningful: it implies that if a farmer expects a wheat yield of 4.0 t/ha, the model's prediction will be off by less than 0.5 t/ha on average — within the range of normal seasonal variation and therefore actionable for planning purposes.

6.2 Regression Coefficients and Feature Significance

Feature	Coefficient	Std. Error	p-value	Significance
Intercept	-1.824	0.412	< 0.001	***
Rainfall (mm)	+0.0042	0.0003	< 0.001	***
Fertilizer (kg/ha)	+0.0198	0.0025	< 0.001	***
Temperature (°C)	-0.0561	0.0071	< 0.001	***
Soil pH	+0.3120	0.0540	< 0.001	***
Pesticide (kg/ha)	+0.0870	0.0210	< 0.01	**
Area (ha) [log]	+0.1430	0.0320	< 0.001	***

All six features were statistically significant ($p < 0.01$). Key findings from the coefficient table:

- Fertilizer had the largest positive effect among controllable inputs (+0.0198 per unit), confirming that optimised fertilizer use is the primary agronomic lever for yield improvement.
- Rainfall contributed meaningfully (+0.0042 per mm), particularly for rainfed crops. A 100 mm increase in annual rainfall corresponds to an expected yield increase of approximately 0.42 t/ha, all else equal.

- Temperature had a negative coefficient (-0.0561 per $^{\circ}\text{C}$), consistent with the thermal stress research: higher average temperatures during critical growth stages reduce yield, especially for wheat and pulses.
- Soil pH showed a strong positive coefficient ($+0.3120$), highlighting that alkaline soils approaching neutral pH (6.5–7.5) are significantly more productive than acidic soils.

6.3 Residual Analysis

The residuals vs. fitted plot showed that residuals were approximately randomly scattered around zero, with no obvious funnel pattern — suggesting homoscedasticity (constant variance) was largely satisfied. The Q-Q plot showed that residuals followed a near-normal distribution with mild departures at the tails, likely due to extreme weather events and unusual soil conditions not captured by the model. These tail behaviours could be addressed in future work by using polynomial regression or a Random Forest regressor.

6.4 Strengths and Limitations

Strengths:

- Highly interpretable — every coefficient has a clear agronomic interpretation.
- Fast training and inference — suitable for deployment on resource-constrained devices in rural areas.
- Strong baseline R^2 of 0.82 achievable with only six features and no hyperparameter tuning.
- Generalises across crops and states when categorical features are included.

Limitations:

- Assumes a linear relationship between features and yield — does not capture threshold effects (e.g., heat stress above 34°C) without feature engineering.
- Categorical encoding of crops and states as ordinal integers is a simplification; a one-hot encoding would be more statistically correct but increases dimensionality.
- Model does not account for time-series patterns (year-on-year rainfall trends, soil degradation over successive seasons).
- Synthetic data limits real-world generalizability; retraining on live data from data.gov.in is essential before field deployment.

6.5 Lessons Learned and Challenges

The most significant challenge encountered was deciding how to handle categorical variables. Initially encoding crop types as 1–10 imposed an artificial ordinal relationship (as if rice > wheat > maize). After reading about ordinal vs. nominal encoding, label encoding was retained for the linear model but noted as a key limitation for future non-linear models.

The log transformation of Area was a valuable discovery — without it, the model's residuals showed a strong heteroscedastic pattern. A single line of code (`np.log1p`) resolved this issue and improved R^2 by approximately 0.04 points.

Temperature's negative effect was initially counterintuitive. Researching the agronomy behind this — thermal stress, stomatal closure, reduced photosynthesis — turned this into a genuine learning moment about the domain knowledge required for responsible ML modelling.

7. Conclusion

This project successfully designed and implemented a Multiple Linear Regression model to predict crop yield (tonnes/hectare) across diverse Indian agricultural regions. Working with a dataset of 5,000 records spanning 10 crops, 15 states, and three growing seasons, the model achieved a test R^2 of 0.821, an MAE of 0.43 t/ha, and an RMSE of 0.59 t/ha — performance metrics that place it firmly within the range of practical usefulness for agricultural advisory applications.

The analysis identified fertilizer application rate, annual rainfall, and soil pH as the three most influential predictors of yield, while temperature emerged as a significant inhibitory factor at high values — findings consistent with peer-reviewed agronomic research and thus validating the model's credibility.

Beyond the technical results, this project demonstrated the full AI workflow: from problem framing and data exploration through preprocessing, modelling, evaluation, and interpretation. The experience highlighted that data quality (handling outliers, transforming skewed features) and domain understanding (why temperature is negative, why pH matters) are as important as model choice in producing trustworthy predictions.

Future extensions of this work could include: (1) incorporating polynomial or interaction terms to capture non-linear temperature effects; (2) using an ensemble model such as Random Forest to improve prediction accuracy; (3) integrating live weather API data from the India Meteorological Department for real-time advisory use; and (4) developing a mobile-friendly interface that allows farmers to enter their local conditions and receive an instant yield estimate with a confidence interval.

Ultimately, this project reinforces the conviction that AI — even in its most foundational form — holds transformative potential for Indian agriculture: making data actionable, making expertise accessible, and making farming more resilient.

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Appendix

Appendix A — Sample Dataset Rows

State	Crop	Season	Rain (mm)	Fert	Temp	pH	Yield
Punjab	Wheat	Rabi	620	180	22.1	7.2	4.81
Maharashtra	Cotton	Kharif	870	95	31.4	6.8	1.93
UP	Rice	Kharif	1100	210	28.9	6.5	3.47
Karnataka	Maize	Rabi	740	140	25.2	6.9	3.12
MP	Soybean	Kharif	960	120	29.6	6.7	1.68

Appendix B — Key EDA Observations (Textual Summary)

- Distribution of Yield: Approximately normal after outlier removal. Mean yield ≈ 3.1 t/ha, SD ≈ 1.4 t/ha.
- Rainfall Distribution: Right-skewed (range 400–2800 mm). States in the northeast and along the Western Ghats show highest rainfall.
- Fertilizer Usage: Ranges from 30–350 kg/ha. Punjab and Haryana have the highest average fertilizer use, consistent with Green Revolution legacy.
- Soil pH: Bimodal distribution — one cluster around 5.5–6.0 (acidic soils, NE India) and another at 7.0–7.8 (alluvial plains of UP, Bihar).

Appendix C — How to Run the Code

15. Open Google Colab (<https://colab.research.google.com>).
16. Upload the dataset CSV file using the Files panel on the left sidebar.
17. Create a new notebook and paste each code block from Section 5 in separate cells.
18. Run all cells using Runtime > Run all.
19. All plots and metric outputs will appear inline in the notebook.
20. Save a copy to Google Drive using File > Save a copy in Drive.

Appendix D — Glossary

Multiple Linear Regression (MLR): A supervised learning algorithm that models the relationship between a dependent variable and two or more independent variables using a linear equation.

R^2 (Coefficient of Determination): A statistical measure from 0 to 1 indicating the proportion of variance in the dependent variable explained by the model. Higher is better.

MAE (Mean Absolute Error): The average of absolute differences between predicted and actual values. Lower is better.

RMSE (Root Mean Squared Error): Square root of the average squared differences between predicted and actual values. Penalises large errors more than MAE.

VIF (Variance Inflation Factor): A measure of multicollinearity. $VIF > 10$ indicates problematic collinearity between predictors.

Kharif Season: Summer cropping season in India (June–November), dependent on the Southwest Monsoon. Major crops: rice, maize, cotton.

Rabi Season: Winter cropping season in India (November–April). Major crops: wheat, mustard, chickpea.

Standardisation (Z-score): Transforming features to have zero mean and unit standard deviation:
 $z = (x - \mu) / \sigma$.